

Homework 2

*Instructor: Henry Lam**Due: Sept 26, 2025*

Problem 1: Inconsistent weather stations. Every evening, two weather stations issue weather forecast for the next day. The weather forecasts are independent. On average, the weather forecast of station 1 is correct 90% of the cases, irrespective of weather type. This percentage is 80% for station 2. Suppose that, without any forecast, the belief is that the next day rains or has sunny weather each with a 50% chance. On a given day, station 1 predicts sunny weather for the next day, whereas station 2 predicts rain. What is the probability that the weather forecast of station 1 will be correct?

Solution:

According to Bayes' rule,

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad (1)$$

we can write,

$$\begin{aligned} & \mathbb{P}(\text{Station 1 is correct} | \text{Station 1 predicts a sunny day, Station 2 predicts a rainy day}) \\ &= \frac{\mathbb{P}(\text{Next day is a sunny day, Station 1 predicts a sunny day and Station 2 predicts a rainy day})}{\mathbb{P}(\text{Station 1 predicts a sunny day, Station 2 predicts a rainy day})}. \end{aligned} \quad (2)$$

To continue, we have

$$\begin{aligned} & \mathbb{P}(\text{Next day is a sunny day, Station 1 predicts a sunny day and Station 2 predicts a rainy day}) \\ &= \mathbb{P}(\text{Next day is a sunny day}) \cdot \mathbb{P}(\text{Station 1 predicts a sunny day} | \text{Next day is a sunny day}) \\ & \quad \cdot \mathbb{P}(\text{Station 2 predicts a rainy day} | \text{Next day is a sunny day, Station 1 predicts a sunny day}). \end{aligned} \quad (3)$$

We know that

$$\mathbb{P}(\text{Next day is a sunny day}) = 0.5, \quad (4)$$

since “the belief is that the next day rains or has sunny weather each with a 50% chance”. We also know that

$$\mathbb{P}(\text{Station 1 predicts a sunny day} | \text{Next day is a sunny day}) = 0.9, \quad (5)$$

since “the weather forecast of station 1 is correct 90% of the cases, irrespective of weather type.” Finally, we know

$$\mathbb{P}(\text{Station 2 predicts a rainy day} | \text{Next day is a sunny day, Station 1 predicts a sunny day}) = 0.2, \quad (6)$$

since “the weather forecast of station 2 is correct 80% of the cases, irrespective of weather type” and “the weather forecasts of two stations are independent”.

Thus, the numerator of (2) equals

$$\begin{aligned} & \mathbb{P}(\text{Next day is a sunny day, Station 1 predicts a sunny day and Station 2 predicts a rainy day}) \\ &= 0.5 \cdot 0.9 \cdot 0.2. \end{aligned} \quad (7)$$

Similarly, we can compute

$$\begin{aligned} & \mathbb{P}(\text{Next day is a rainy day, Station 1 predicts a sunny day and Station 2 predicts a rainy day}) \\ &= 0.5 \cdot 0.8 \cdot 0.1. \end{aligned} \quad (8)$$

However, the denominator of (2) is

$$\begin{aligned}
 & \mathbb{P}(\text{Station 1 predicts a sunny day and Station 2 predicts a rainy day}) \\
 &= \mathbb{P}(\text{Next day is a sunny day, Station 1 predicts a sunny day and Station 2 predicts a rainy day}) \\
 & \quad + \mathbb{P}(\text{Next day is a rainy day, Station 1 predicts a sunny day and Station 2 predicts a rainy day}) \\
 &= 0.5 \cdot 0.9 \cdot 0.2 + 0.5 \cdot 0.8 \cdot 0.1.
 \end{aligned} \tag{9}$$

Thus, (2) is

$$\begin{aligned}
 & \frac{\mathbb{P}(\text{Next day is a sunny day, Station 1 predicts a sunny day and Station 2 predicts a rainy day})}{\mathbb{P}(\text{Station 1 predicts a sunny day, Station 2 predicts a rainy day})} \\
 &= \frac{0.5 \cdot 0.9 \cdot 0.2}{0.5 \cdot 0.9 \cdot 0.2 + 0.5 \cdot 0.8 \cdot 0.1} \\
 &= \frac{9}{13}.
 \end{aligned} \tag{10}$$

Problem 2 *USA Today* reported that “high-volume” hospitals performed 77% of bladder removal surgeries; “low-volume” hospitals performed 23%. In the first two weeks after surgery, 3.1% of patients at low-volume centers died, compared to 0.7% at the high-volume centers.

a) Compute the probability that a randomly chosen bladder-cancer patient had surgery at a high-volume hospital and survived the first two weeks after surgery.

b) Calculate the probability that a randomly chosen bladder-cancer patient survived the first two weeks after surgery.

c) If two bladder-cancer patients were chosen randomly, determine the probability that only one would survive the first two weeks after surgery.

d) If three bladder-cancer patients were chosen randomly, determine the probability that at least two would survive the first two weeks after surgery.

Solution:

Let’s define the following events:

- H : High volume hospital
- H^c : Low volume hospital
- D : dies two weeks after surgery
- D^c : lives after two weeks of surgery

Then, the data provided can be written as:

- $P(H) = 0.77$
- $P(H^c) = 0.23$
- $P(D|H) = 0.7\%$
- $P(D|H^c) = 3.1\%$
- (a) We are looking for the probability of H and D^c

$$\begin{aligned}
P(D^c \cap H) &= P(D^c|H) \cdot P(H) \\
&= (1 - P(D|H)) \cdot P(H) \\
&= \left(1 - \frac{0.7}{100}\right) \cdot 0.77 \approx 0.765
\end{aligned}$$

- (b) Using the law of total probability:

$$\begin{aligned}
P(D^c) &= P(D^c \cap H) + P(D^c \cap H^c) \\
&= P(D^c|H) \cdot P(H) + P(D^c|H^c) \cdot P(H^c) \\
&= (1 - P(D|H)) \cdot P(H) + (1 - P(D|H^c)) \cdot P(H^c) \\
&= \left(1 - \frac{0.7}{100}\right) \cdot 0.77 + \left(1 - \frac{3.1}{100}\right) \cdot 0.23 \\
&\approx 0.987
\end{aligned}$$

- (c) As the chosen patients are independent, we are looking for the event that either the first one lives and the second dies or the other way around. Let's call this event C :

$$\begin{aligned}
P(C) &= P(D_1 \cap D_2^c) + P(D_2 \cap D_1^c) \\
&= 2 \cdot P(D) \cdot P(D^c) \\
&= 2 \cdot (1 - P(D^c)) \cdot P(D^c) \\
&\approx 0.025
\end{aligned}$$

- (d) We can divide this event E in: exactly two people survive (F) and all three people survive (G)

$$\begin{aligned}
P(E) &= P(F) + P(G) \\
P(F) &= P(D_1^c \cap D_2^c \cap D_3) + P(D_1^c \cap D_2 \cap D_3^c) + P(D_1 \cap D_2^c \cap D_3^c) \\
P(F) &= 3 \cdot [P(D^c)]^2 \cdot P(D) \approx 0.037 \\
P(G) &= [P(D^c)]^3 \approx 0.963 \\
\rightarrow P(E) &= 0.9995
\end{aligned}$$

Problem 3 This year experts project that 20% of all taxpayers will file an incorrect tax return. The Internal Revenue Service (IRS) itself is not perfect. IRS auditors claim that there is an error when no problem exists about 10% of the time. The audits also indicate no error with a tax return when in fact there really is a problem about 30% of the time.

The IRS has just notified a taxpayer there is an error in his return. What is the probability that the return actually has an error?

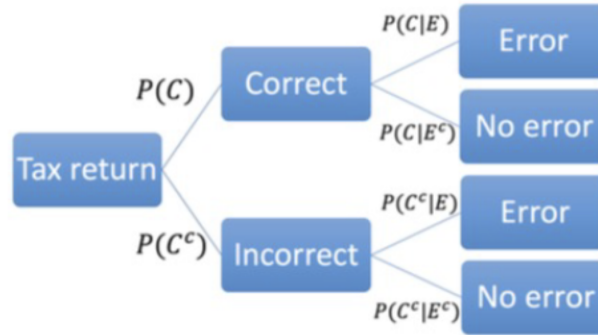
Solution:

If we name C the event of the tax return being correct and E the event of getting an error, then we know that:

- $P(C) = 0.8$ $P(C^c) = 0.2$

- $P(E|C) = 0.1$ $P(E^c|C) = 0.9$
- $P(E|C^c) = 0.7$ $P(E^c|C^c) = 0.3$

If we consider the following chance tree:



$$\begin{aligned}
 P(C^c|E) &= P(E \cap C^c)/P(E) \\
 &= \frac{P(E|C^c) \cdot P(C^c)}{P(E \cap C) + P(E \cap C^c)} \\
 &= \frac{P(E|C^c) \cdot P(C^c)}{P(E|C) \cdot P(C) + P(E|C^c) \cdot P(C^c)} \\
 &= \frac{0.7 \cdot 0.2}{0.1 \cdot 0.8 + 0.7 \cdot 0.2} \approx 0.6364
 \end{aligned}$$

An alternative method (chance tree):

$$\begin{aligned}
 P(C^c|E) &= P(E \cap C^c)/P(E) \\
 &= \frac{P(E|C^c) \cdot P(C^c)}{P(E \cap C) + P(E \cap C^c)} \\
 &= \frac{P(E|C^c) \cdot P(C^c)}{P(E|C) \cdot P(C) + P(E|C^c) \cdot P(C^c)} \\
 &= \frac{0.7 \cdot 0.2}{0.1 \cdot 0.8 + 0.7 \cdot 0.2} \approx 0.6364
 \end{aligned}$$

Problem 4. On independent events:

- Consider two independent events A and B. Prove that A and B^c are also independent.
- Prove that if $E_1, E_2, \dots, E_n$ are independent events, then

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = 1 - \prod_{i=1}^n [1 - P(E_i)]$$

solutions:

(a)

If A and B^c are independent, then $P(A \cap B^c) = P(A) \cdot P(B^c)$. So we need to prove this using that A and B are independent (This means: $P(A \cap B) = P(A) \cdot P(B)$) Using the law of total probability:

$$A = (A \cap B) \cup (A \cap B^c)$$

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap B^c) && \text{Using that } (A \cap B) \text{ and } (A \cap B^c) \text{ are mutually exclusive} \\ \rightarrow P(A \cap B^c) &= P(A) - P(A \cap B) && \text{by using the independence of A and B} \\ &= P(A)[1 - P(B)] = P(A) \cdot P(B^c) \end{aligned}$$

(b)

$$\begin{aligned} P(E_1 \cup E_2 \cup \dots \cup E_n) &= 1 - P((E_1 \cup E_2 \cup \dots \cup E_n)^c) \\ &= 1 - P(E_1^c \cap E_2^c \cap \dots \cap E_n^c) \\ &= 1 - P(E_1^c) \cdot P(E_2^c) \cdot \dots \cdot P(E_n^c) \\ &= 1 - \prod_{i=1}^n [P(E_i^c)] \\ &= 1 - \prod_{i=1}^n [1 - P(E_i)] \end{aligned}$$

Problem 5. Alice is hired by CVS to improve its overall product display optimization in its New York stores. For some reason, she begins her work with flashlights in the electrical supplies section. She finds out that in the display of electrical supplies section in each store in New York, 20% of the flashlights are of type 1, 30% are of type 2 and 50% are of type 3. In addition, she learns that the probability that Type 1 flashlight will give over 100 hours of use is 0.7, with the corresponding probabilities for Type 2 and Type 3 flashlights being 0.4 and 0.3 respectively.

a) To model customer buying behaviour, Alice assumes that any customer who wants to buy a flashlight chooses one uniformly at random from the display. Under this assumption, what is the probability that a randomly chosen flashlight that a customer buys will give more than 100 hours of use?

b) Given the flashlight lasted over 100 hours, what is the conditional probability that it was a type j flashlight, with $j = 1, 2, 3$

Solution:

Defining H as the event of lasting over 100 hours and A, B, C the events that the flashlights are type 1, 2 and 3. Then, the information provided is:

- $P(A) = 0.2 \quad P(H | A) = 0.7$
- $P(B) = 0.3 \quad P(H | B) = 0.4$
- $P(C) = 0.5 \quad P(H | C) = 0.3$

(a) We are looking for the value of $P(H)$ by using the law of total probability:

$$\begin{aligned} P(H) &= P(H | A) \cdot P(A) + P(H | B) \cdot P(B) + P(H | C) \cdot P(C) \\ &= 0.7 \cdot 0.2 + 0.4 \cdot 0.3 + 0.3 \cdot 0.5 = 0.41 \end{aligned}$$

(b) We are looking for the values of $P(A | H), P(B | H), P(C | H)$: We know that:

$$P(A \cap H) = P(H | A) \cdot P(A) \rightarrow P(A | H) = \frac{P(H | A) \cdot P(A)}{P(H)}$$

We can use the same equation replacing A for B, C . Then, using previous results:

$$P(A | H) = \frac{0.7 \cdot 0.2}{0.41} \approx 0.342$$

$$P(B | H) = \frac{0.4 \cdot 0.3}{0.41} \approx 0.293$$

$$P(C | H) = \frac{0.3 \cdot 0.5}{0.41} \approx 0.366$$

Problem 6. A pair of fair dice is rolled. Let E denote the event that the sum of the dice is equal to 7

- (a) Show that E is independent of the event that the first die lands on 4
 (b) Show that E is independent of the event that the second die lands on 3 .

Solution:

- (a). Denote $F = \{ \text{The first die lands on 4} \}$,

$$P(E \cap F) = P(\text{The first die lands on 4, the second die lands on 3}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

$$P(E) = \frac{6}{36} = \frac{1}{6}$$

$$P(F) = \frac{1}{6}$$

$$\Rightarrow P(E \cap F) = P(E) \cdot P(F)$$

- (b). Denote $G = \{ \text{The second die lands on 3} \}$,

$$P(E \cap G) = P(\text{The first die lands on 4, the second die lands on 3}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

$$P(E) = \frac{6}{36} = \frac{1}{6}$$

$$P(G) = \frac{1}{6}$$

$$\Rightarrow P(E \cap G) = P(E) \cdot P(G)$$

Problem 7. Let A, B, C be events such that $P(A) = 0.2, P(B) = 0.3, P(C) = 0.4$ Find the probability that at least one of the events A and B occurs if

- (a) A and B are mutually exclusive;
 (b) A and B are independent.

Find the probability that all of the events A, B, C occurs if

- (c) A, B, C are independent;
 (d) A, B, C are mutually exclusive.

Solution:

- (a).

$$P(A \cup B) = P(A) + P(B) = 0.5$$

- (b).

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A) \cdot P(B) = 0.44$$

- (c).

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C) = 0.024$$

- (d).

$$P(A \cap B \cap C) = 0$$